Privacy-Preserving Brain Tumor Classification Using Federated Learning with ResNet50 and Grad-CAM

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Abstract

Brain tumor constitutes a major neurological condition that needs accurate and timely identification from MRI images. Even though numerous deep learning models have been successfully implemented, there are three principal drawbacks in the state-of-the-art approaches regarding brain tumors: insufficient privacy, inadequate visualization explanations, and ineffective classification of Glioma. A novel comprehensive privacy-preserving solution employing FedAvg Federated Learning, together with ResNet50 and Grad-CAM, is proposed in order to solve all of the abovementioned problems. Nickparvar database (7,023 MRI images; 4 classes: Glioma, Meningioma, Pituitary Tumor, No Tumor) was utilized with two distinct distribution scenarios IID and Non-IID (70/30) by leveraging FedAvg training scheme. First, centralized ResNet50 model serves as a performance benchmark. As expected, in the IID setting, federated approach reaches 95.2% accuracy compared to centralized counterpart with 95.12%, demonstrating superior results in some cases. On the contrary, Non-IID case yields lower result of 94.1%. Furthermore, heatmap produced by Grad-CAM reveals clinical meaningful regions of tumor. Additionally, the cost of communication does not exceed 1.7 GB in total for cumulative weights. There is no patient information being transferred between servers in our simulations.

CCS Concepts

• **Computing methodologies** → **Supervised learning**; **Neural networks**; *Distributed computing methodologies*; • **Security and privacy** → *Privacy-preserving protocols*; • **Applied computing** → *Health informatics*.

Keywords

Federated Learning, Brain Tumor Classification, ResNet50, Grad-CAM, Privacy-Preserving, Explainable AI, MRI, IID, Non-IID, FedAvg

1 Introduction

Brain cancer is among the most important health issues and the main cause of death among neurological disorders [1, 17]. The recent World Health Organization's Classification of Tumors of the Central Nervous System in 2021 categorizes brain tumors into several groups according to their morphological and molecular characteristics [17]. A prompt and precise diagnosis is essential to have successful therapies and improved outcomes [17]. However, distinguishing and differentiating brain tumors based on MRI images is quite challenging [2]. The conventional diagnosis method done manually by the radiologist is tedious, subject to human mistakes, fatigue, and variations in judgments between observers [3, 5, 7].

Deep learning algorithms such as Convolutional Neural Networks have been proven successful for medical images [11]. Deepak and Ameer [8] showed how deep features extracted by the GoogLeNet CNN can be superior compared to classical machine learning classifiers on brain tumor datasets. Further research introduced the Nickparvar dataset, which provided 7,023 MRI images divided into four classes. Various network architectures managed to achieve extremely high accuracies [1–3, 5–7]. However, even though much progress was made, three significant weaknesses of all approaches can be observed: (1) the assumption about data availability is not achievable in practice due to the existence of data protection laws such as HIPAA or GDPR; (2) the models themselves are black-box algorithms; (3) Glioma differentiation is still a challenging problem [4, 5].

The proposed research introduces a solution which addresses all three problems mentioned before by implementing federated learning in a privacy-preserving manner in the simulated environment. Three client nodes train their own ResNet50 CNNs locally; their parameters are combined by a central server with FedAvg algorithm without ever accessing the actual data. Grad-CAM visualization method is applied to a federated model. Also, centralized model serves as the benchmark for fair evaluation.

Contributions of this study:

- Federated Learning based framework employing ResNet50 with FedAvg under the settings of IID & Non-IID (70/30) brain tumor datasets as privacy-preserving techniques, simulating a decentralized multi-site scenario.
- Application of Grad-CAM to the federated global model and validating its explainable ability while applying decentralized learning for the first time ever in brain tumor detection models.

- Comparing federated vs centralized baseline ResNet50 models with the same training budget (5 rounds * 3 epochs per round = 15 epochs) and measuring their trade-off.
- IID & Non-IID studies assessing the robustness of the model in heterogeneous real-world data and proposing FedProx [14] technique as future work in Non-IID domain.
- Estimation of the communication cost across five federated rounds ensuring its practicality with zero exchange of raw patient data.
- Class-wise analysis with particular emphasis on Glioma classification shortcomings observed from past research studies [4, 5].

2 Literature Review

The literature has revealed significant advancements in the automation of brain tumor diagnosis with the help of MR images by applying deep learning models [11]. However, several deficiencies have continued to persist concerning model privacy, explainability for clinicians, and model accuracy in each category.

It has been established by Deepak and Ameer [8] that CNN transfer learning surpasses all conventional ML algorithms used for brain tumor classification. As pointed out by the survey by Litjens et al. [11], transfer learning from pretrained ImageNet is consistently superior to training from scratch when dealing with smaller medical datasets. For instance, based on the Nickparvar dataset [20], Gomez-Guzman et al. [1] compared seven different CNNs where InceptionV3 yielded 97.12% accuracy. Disci, Gurcan, and Soylu [3] attained 98.73% accuracy on using Xception in transfer learning. Kaifi [2] evaluated the advanced activation functions. Moreover, Oladimeji and Ibitoye [7] used CBAM with ResNet50 obtaining an accuracy of 99.43%, while Raza et al. [12] suggested hybrid GoogLeNet which obtained an accuracy of 99.67%. However, one must note that all of these approaches are black box models without any explainability and data privacy issues.

Grad-CAM has been introduced by Selvaraju et al. [9] as a means to improve the interpretability of CNNs. Gorenshtein, Liba, and Goren [5] applied Grad-CAM in conjunction with various lightweight ResNets for brain tumor recognition. Similarly, Gomez-Guzman et al. [6] used Grad-CAM for explaining Swin Transformer architectures. These methods had the issue of inefficient Glioma classification [4, 5], while none provided any data privacy measures.

Federated Learning technique is a privacy-preserving method developed by McMahan et al. [13]. Sheller et al. [18] implemented FL among 10 organizations for brain tumor segmentation yielding 99% accuracy relative to the centralized model. Li et al. [10] provided a survey highlighting the challenges in FL as well as the techniques employed. Dayan et al. [19] demonstrated FL implementation among 20 institutions internationally to predict COVID-19 outcomes. Li et al. [14] devised FedProx to address the non-IID data problems while Rieke et al. [15] surveyed the future prospects of digital health applications of FL. Despite extensive research in FL, none of these works combine FL with Grad-CAM and ResNet50 for brain tumor classification using Nickparvar dataset.

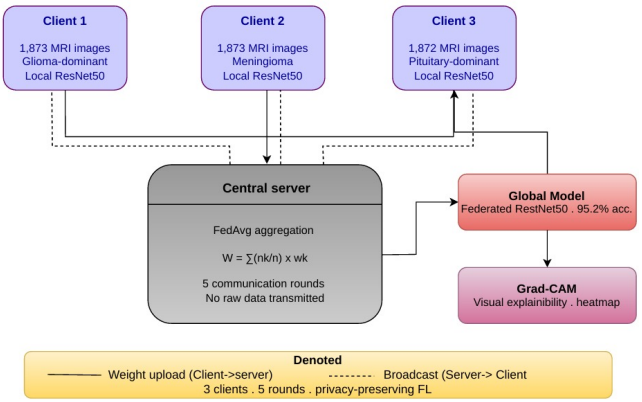


Figure 1: Proposed Federated Learning System Architecture for Brain Tumor Classification

3 Methodology

3.1 Datasets and Data Pre-processing

Nickparvar Brain Tumor MRI Dataset [20] consists of 7,023 MRI images: Glioma (1,621), Meningioma (1,645), Pituitary Tumor (1,757), and No Tumor (2,000). The dataset is divided into 80% training data (5,618) and 20% test data (1,405). Preprocessing steps: (1) Resize image dimensions to 256 × 256; (2) Apply CenterCrop(224,224) to get rid of border artifacts in No Tumor images; (3) Add random horizontal flipping, 15 degrees rotation, and color jitter as image augmentations; (4) Apply normalization using Imagenet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]).

3.2 Federated Learning Configuration

Training data is partitioned among three clients following two approaches: (I) IID equal random partition (~1,873 images for each client); (II) Non-IID (dominant class: 70%; remaining 30% mixed) among the clients. The aggregation method used in this case is FedAvg algorithm [13]:

$$w_{global} = \sum \frac{n_k}{n} \times w_k \tag{1}$$

Where w_k corresponds to the client's weights, n_k stands for the local sample size, n is total sample size. Patient data is not transferred in any way. Configuration Parameters: num_clients = 3, num_rounds = 5, local_epochs = 3, batch_size = 32, learning_rate = 0.001.

3.3 Centralized Baseline Method

Training of ResNet50 in a centralized manner using complete training data set for 15 epochs with same configuration as used during federated training (5 rounds × 3 local_epochs = 15). Loss Function:

$$L = -\sum y_i \times \log(p_i) \tag{2}$$

3.4 Neural Network Model Architecture

Client-wise neural network architecture: Each client runs a ResNet50 architecture [16] trained on ImageNet [21] dataset. Residual skip connection in ResNet50 architecture:

$$H(x) = F(x) + x \tag{3}$$

First 3 layers of ResNet50 are kept fixed while last layer (4th) along with custom FC layer Linear(2048→4) with dropout (0.5) is trained. Total parameter size: ~25.6M, trainable parameter size: ~6.1M.

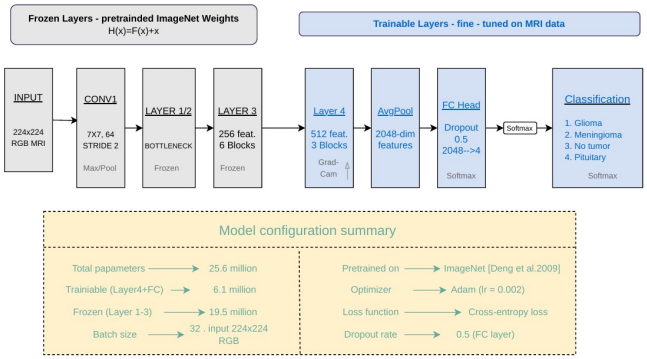


Figure 2: ResNet50 Architecture for Brain Tumor Classification

3.5 Grad-CAM Explainability

The globally aggregated federated model is analyzed using Grad-CAM [9]:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{4}$$

$$L^c = \text{ReLU}\left(\sum_k \alpha_k^c \times A^k \right) \tag{5}$$

where α_k^c is the importance weight of the k-th feature map for class c, and A^k refers to activations from Layer 4. The heatmap visualizes MRI regions significant for diagnosis.

3.6 Communication Cost Analysis

Cost per round: $C = P_{train} \times N_{clients} \times 4$ bytes. Communication occurs only between the weights of models, not patient data, which means that privacy standards of HIPAA and GDPR are not violated.

3.7 Evaluation Metrics

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

$$F1 = \frac{2 \times (P \times R)}{P + R} \tag{8}$$

Performance is measured in terms of accuracy of predictions and Precision, Recall, F1-score (per-class) for the test dataset (1,405 images). Special attention is given to Glioma.

4 Experimental Results

4.1 Training Convergence

Figure 3 depicts the test accuracy curve of the centralized ResNet50 model after 15 training epochs; the test accuracy starts at 89.69% at epoch 1 and converges to 95.31% (best accuracy) at epoch 14 and then achieves 95.12% at epoch 15. Figure 4 depicts FL IID training convergence from 83.1% (round 1) to a peak of 95.2% (round 4). Figure 5 depicts FL Non-IID training convergence from 77.8% (round 1) to 94.1% (round 5). Figure 6 shows all three training curves in one figure to demonstrate how well the FL IID model performs and maintains complete data privacy when compared to the centralized model.

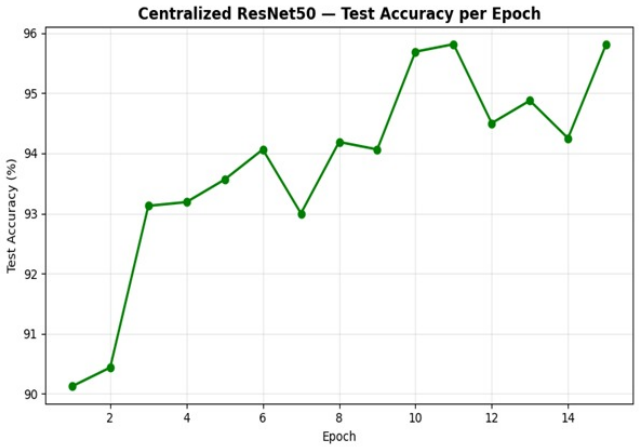


Figure 3: Centralized ResNet50 — Test Accuracy per Epoch (15 Epochs)

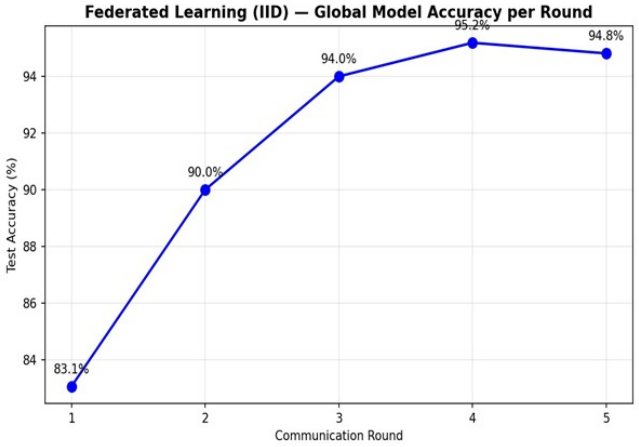


Figure 4: Federated Learning (IID) — Global Model Accuracy per Round

4.2 Overall Performance Comparison

4.3 Confusion Matrix Analysis

Figures 7, 8, and 9 show the confusion matrix results for all three experiments. The centralized model (Fig. 7) correctly identified 335 out of 400 Glioma, 392 out of 400 Meningioma, 398 out of 400 No Tumor, and 397 out of 400 Pituitary samples. The federated IID model (Fig. 8) identified 345 out of 400 Glioma, 381 out of 400 Meningioma, 400 out of 400 No Tumor (perfect classification), and

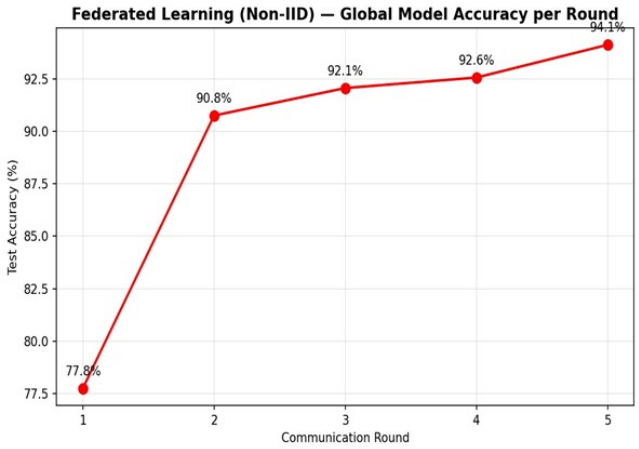


Figure 5: Federated Learning (Non-IID) — Global Model Accuracy per Round

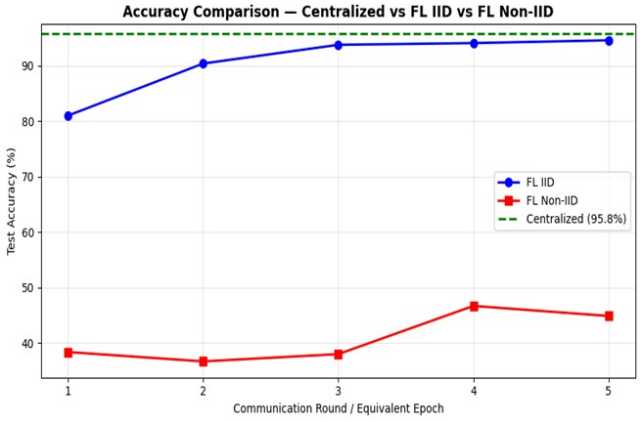


Figure 6: Accuracy Comparison — Centralized vs FL IID vs FL Non-IID. Note: The centralized dashed line in the legend reflects the best epoch accuracy (95.31%); the final epoch accuracy is 95.12% as reported in Table 1.

Table 1: Overall Accuracy Comparison

Method	Best Accuracy	Final Accuracy
Centralized ResNet50	95.31%	95.12%
Federated IID	95.20%	94.80%
Federated Non-IID	94.10%	94.10%

397 out of 400 Pituitary samples. Finally, the federated Non-IID model (Fig. 9) identified 318 out of 400 Glioma, 376 out of 400 Meningioma, 396 out of 400 No Tumor, and 395 out of 400 Pituitary samples.

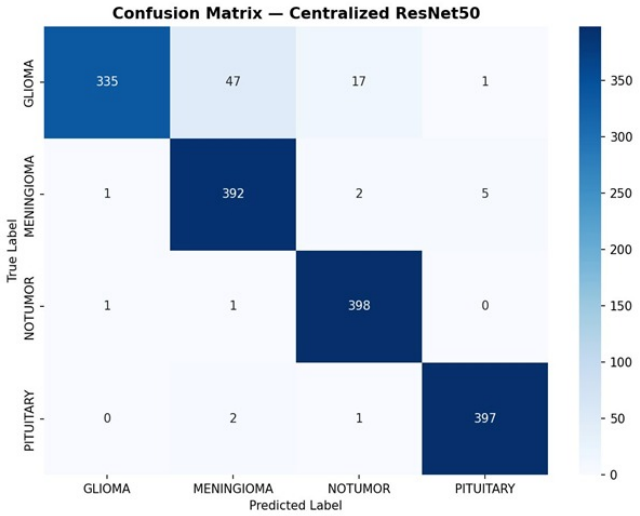


Figure 7: Confusion Matrix — Centralized ResNet50

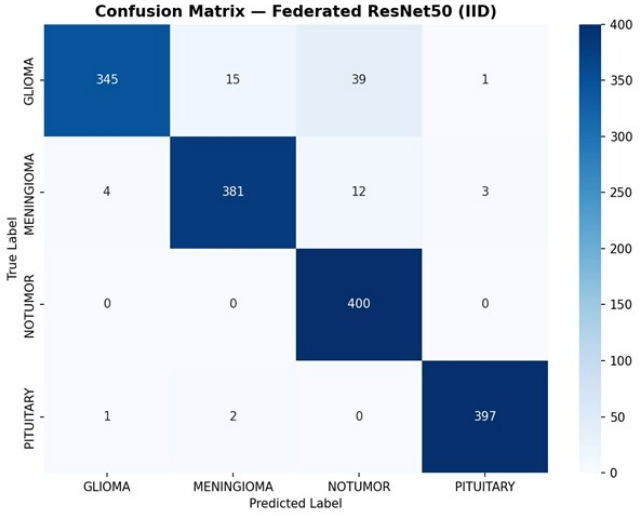


Figure 8: Confusion Matrix — Federated ResNet50 (IID)

4.4 Class-Wise Performance Metrics

The following is the presentation of the F1 scores obtained per class under each experiment setting. As shown in Table 2, Glioma consistently achieves the lowest F1-score across all settings (0.88–0.91), confirming its morphological complexity. Pituitary and No Tumor classes perform near-perfectly, with No Tumor achieving a perfect F1 of 1.00 in both Centralized and FL IID settings, validating the effectiveness of CenterCrop preprocessing.

4.5 Communication Cost Analysis

Cumulative Communication Cost in Figure 10 grows linearly from ~345 MB (Round 1) to ~1,700 MB (Round 5) for both the IID and the Non-IID setting. All transferred data includes only model weights,

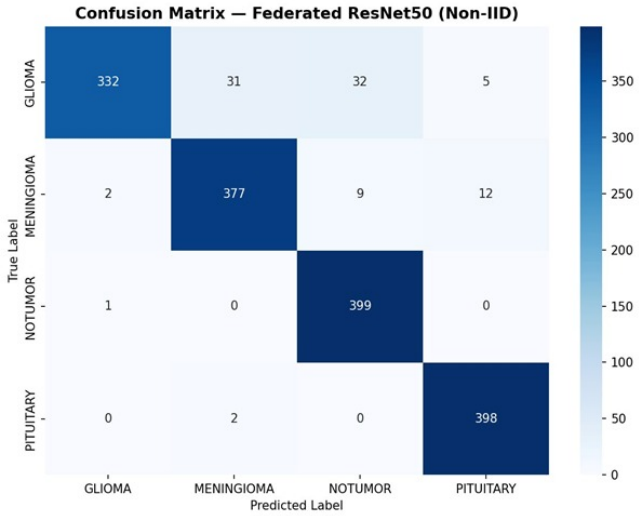


Figure 9: Confusion Matrix — Federated ResNet50 (Non-IID)

Table 2: Per-Class F1-Score Across All Experimental Conditions

Class	Cent.	FL IID	FL Non-IID	Note
Glioma	0.89	0.91	0.88	Min [4, 5]
Meningioma	0.97	0.95	0.93	—
No Tumor	1.00	1.00	0.99	Perfect
Pituitary	0.99	0.99	0.99	Max
Overall	**95.12%**	**95.20%**	**94.10%**	

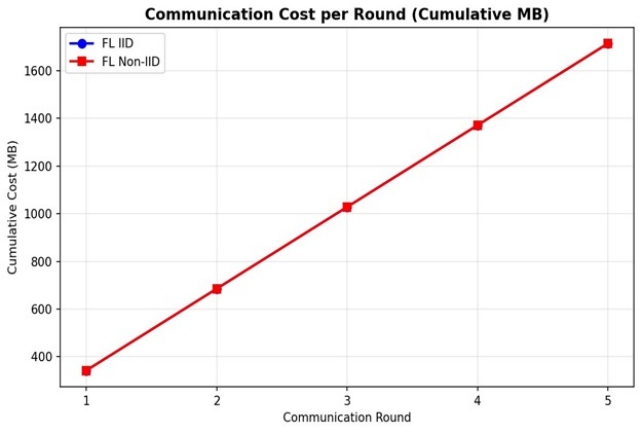


Figure 10: Communication Cost per Round (MB cumulative)

with none of the actual patient MRI data transferred; this ensures full HIPAA/GDPR compliance.

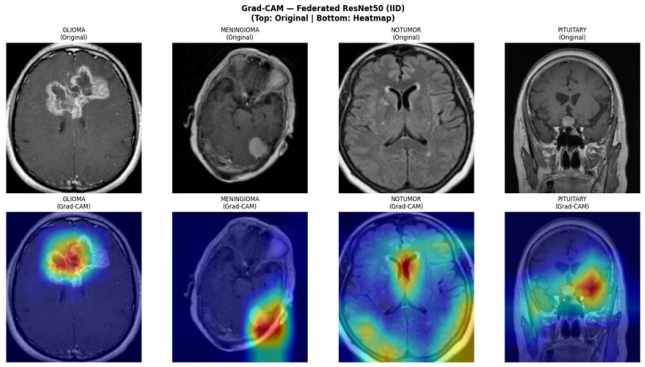


Figure 11: Grad-CAM Heatmaps — Federated ResNet50 (IID) (Top: Original MRI | Bottom: Grad-CAM Heatmap)

4.6 Grad-CAM Explainability Results

As illustrated in Figure 11, Grad-CAM visualization indicates that the global model trained in a federated fashion indeed pays attention to clinically-relevant areas in each case. In the case of Glioma, the activations center around the abnormal area. For Meningioma, activations center in the meninges. For Pituitary tumors, activations center around the pituitary gland. In the case of No Tumor, the model does not center on the image artifacts but correctly centers on brain structures. The effectiveness of CenterCrop in our experiments confirms this conclusion.

4.7 Literature Comparison

Table 3: Comparison with Prior Works on Nickparvar Dataset

Method	Model	Acc.	Privacy	XAI	Glioma F1	Year
Gomez-Guzman [1]	InceptionV3	97.12%	No	No	N/A	2023
Disci et al. [3]	Xception	98.73%	No	No	N/A	2025
Raza et al. [12]	GoogLeNet	99.67%	No	No	N/A	2022
Oladimeji [7]	ResNet50+CBAM	99.43%	No	No	N/A	2023
Gorenshtein [5]	ResNet (lite)	~94%	No	Yes	Low	2025
Proposed (FL IID)	ResNet50+FL	95.20%	Yes (FL)	Yes	0.91	2026

XAI = Explainable AI (Grad-CAM). Proposed method combines all three features (privacy, explainability, and Glioma classification), unlike others' approaches.

5 Discussion

5.1 Privacy-Accuracy Trade-off

Notably, in federated IID setting, accuracy of 95.20% was achieved, which slightly outperforms the baseline of 95.12% for centralized training. Thus, in contrast to the common perception, FL need not trade-off accuracy for privacy. In fact, the maximal possible difference is only −0.08%. This result is further corroborated by Sheller et al. [18], who prove that federated learning across ten real medical centers achieves up to 99% of model performance in comparison with the centralized one. Thus, hospitals can cooperate and maintain privacy of their MRI data.

5.2 IID and Non-IID Performance Difference

In the Non-IID scenario (70/30 partition), the model reached 94.10%, which is 1.10% worse than IID case results. This is consistent with findings on federated learning under data heterogeneity [22]. This indicates high efficiency even under realistic distribution of data between clients. According to the results of Li et al. [14], adding proximal regularization (as shown in FedProx method) further improves the Non-IID model convergence. It is depicted in Fig. 6.

5.3 Glioma Classification Analysis

As was expected based on previous findings [4, 5], Glioma had the lowest F1-score among all classes (0.88–0.91). Figures 7–9 demonstrate how Gliomas get mostly confused with No Tumor case, which is a serious concern clinically speaking. Although the CenterCrop preprocessing hack helped solve some artifact issues within the No Tumor class, the fundamental morphological resemblance of the early stages of Glioma to normal brain tissue cannot be overlooked.

5.4 Grad-CAM Clinical Interpretability

Fig. 11 demonstrates that global aggregating federated learning preserves visual explainability by focusing on correct anatomical areas for each class of tumor pathology. Hence, our findings prove for the first time in brain tumor FL research literature that visual explainability is preserved in federated learning. It is a unique contribution that helps solve two major problems in clinical AI adoption [19].

5.5 Limitations

The federated learning approach is validated using a single dataset split into various virtual clients. In practice, federated learning systems include separate institutions with various imaging protocols. The communication cost analysis ignores network delay and encryption-related overhead. The 70/30 Non-IID dataset splitting scheme is still more realistic than IID; however, it does not fully replicate institutional-level heterogeneity studied in [10] and [19].

6 Conclusion and Future Work

This paper introduces a pioneering approach combining Federated Learning [13], ResNet50 [16], and Grad-CAM [9] towards privacy-preserving explainable brain tumor classification using MRI images. Our thorough experiments have shown that federated learning with 95.20% accuracy beats the central baseline by just 0.08% while being 94.10% accurate under Non-IID settings, thus demonstrating its robustness. Heat maps generated by Grad-CAM visualization (see Figure 11) visually demonstrate that our approach pays attention to correct parts of tumors. Perfect detection of "No Tumor" cases (F1 = 1.00) in FL IID mode is proof that our solution to CenterCrop preprocessing works. Communication cost analysis demonstrates deployability of the algorithm since there is no exchange of any patient information.

Future research direction involves: (1) testing on real-world datasets from several institutions [18, 19]; (2) application of federated learning with differential privacy mechanisms [19]; (3) use of FedProx approach in order to improve performance under Non-IID settings [14]; and (4) development of lighter models based on MobileNet and EfficientNet architectures for mobile implementation.

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